

VOLUME 0

Master Guide and Table of Contents

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-000
Revision	B
Generated	2026-08-07
Source commit	a3730f9
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION

The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Revision B release position

REV B SUPERSEDES REV A
Rev A used a 7i97T-centered design, one 7i84U, incorrect resolver identifiers, retired evidence states, and an unsafe 7i84U TB1 polarity map. Rev B is rebuilt from the live 7i80HDT/7i44/7i49/two-7i84U authority and must be used in its place.

Area	Rev B consolidated position
Host	Mesa 7i80HDT over hm2_eth at 192.168.1.121
P1	7i44; physical channels 0/1 to 7i84U-A/B
P2	Plain 7i49; RES0/1/2 and AOUT0/1/2/3
P3	Empty; no bare 3.3 V GPIO field wiring
Field I/O	Two 7i84Us; 43 DI / 25 DO allocated, 21 DI / 7 DO reserve
7i84U TB1	1/2 VFIELDB, 3/4 VFIELDA, 5 VIN, 6/7/8 common
Firmware	Efinix resolver bitfile name/binary/provenance remain unverified
Evidence	Every physical channel prints its current authority_status

2. Volume map

Vol	Document	Title	Primary audience
0	MZK-LCNC-000	Master Guide and Table of Contents	All users
1	MZK-LCNC-001	Overview and Safety	Owner / safety reviewer
2	MZK-LCNC-002	Operation Manual	Qualified operator
3	MZK-LCNC-003	Mesa Conversion and Installation	Integrator
4	MZK-LCNC-004	Wiring Diagrams and I/O Reference	Electrician / integrator
5	MZK-LCNC-005	Maintenance and Troubleshooting	Maintenance
6	MZK-LCNC-006	Reference, Glossary, and Index	All users

3. Master table of contents

Volume	Sections
V0	Revision position; volume map; source hierarchy; pre-power dashboard; document control
V1	Machine overview; retained/replaced systems; safety authority; LOTO; E-stop; DC bus; Z brake; hazards
V2	Applicability; control station; power-up; homing; jogging; spindle; ATC; G-code; auxiliaries; shutdown; alarms
V3	Architecture; hardware; Ethernet; firmware; resolver interface; field power; HAL/INI; staged bring-up; rejected designs
V4	Five wiring figures; 7i49; 7i44 links; both 7i84U TB1 maps; all 96 digital I/O terminals; conflicts; status legend
V5	Maintenance intervals; resolver/analog/network health; motion/spindle/ATC troubleshooting; rollback; records
V6	OEM manual library; project file map; external sources; glossary; superseded claims; alphabetical index

4. Authority hierarchy

- Electrical channel authority: mesa/current_pin_authority.csv.
- Control logic and binding: linuxcnc/*.hal and mazak_vqc_20_40.ini must agree with the authority CSV.
- Narrative documents interpret the design but cannot override the CSV or HAL.
- These PDFs are release artifacts generated from the source commit printed on every page.
- OEM machine-side wiring remains governed by Mazak 41434WB and the verified as-built cabinet trace.

Authority status	Rows
COMMISSIONING_PENDING	51
DEFERRED	1
HOLD_CONFLICT	4
OPTIONAL_VERIFY	1
PROPOSED	39

Authority status	Rows
SPARE	35
UNBOUND	1

5. Pre-power deliverable dashboard

ID	Deliverable	Planning state
D1	As-built one-line + terminal plan	NOT DRAFTED
D2	Installed nameplate register	PARTIAL (checklists drafted; photos pending)
D3	Immutable Mesa firmware package	NOT ASSEMBLED (bitfile TBD)
D4	I/O checkout sheet	NOT DRAFTED
D5	Hardware E-stop risk assessment + schematic	PARTIAL (software chain drafted)
D6	Shared-bus precharge / discharge procedure	DRAFTED (measurements pending)
D7	Enable / fault / ready / Z-brake timing budget	TEMPLATE DRAFTED (measurements pending)
D8	Resolver phasing / commissioning procedure	PROCEDURE DRAFTED (measurements/traces pending)
D9	First-move plan	PROCEDURE DRAFTED (machine values/signatures pending)
D10	Velocity-mode tuning procedure + saved traces	PROCEDURE DRAFTED (traces pending)
D11	Physical travel / envelope survey	PARTIAL (Y done; X, Z pending)
D12	FR-SX speed / orient / gear state diagram	DIAGRAM DRAFTED (terminal/timing proof pending)
D13	ATC hazard analysis + dry-cycle fixture	NOT DRAFTED
D14	Network / real-time qualification + automatic response	PROCEDURE DRAFTED (logs pending)
D15	Restore / rollback package	TEMPLATE DRAFTED (evidence/photos/ledger pending)
D16	Signed hold points	NOT DRAFTED

NO SIGNED HOLD POINTS

D16 is not drafted. Control power, DC bus, amplifier enable, Z-brake release, spindle rotation, first motion, and automatic M6 remain blocked by their applicable D1-D15 prerequisites.

6. Document control

- Stable filenames are retained for Volumes 0-6 so links do not drift.
- The combined file is revisioned: Mazak_VQC2040_Complete_Manual_RevB.pdf.
- manifest.json records SHA-256 and page count for every released PDF.
- Regenerate with: `uv run scripts/build_manual_set.py`.
- Run `scripts/validate_authority.py` and `scripts/validate_control_logic.py` before release.